

Counterexample-Guided Short-Horizon Shielding of Generative Robot Policies

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Submitted to ICRA 2026 (double-blind)

Abstract—Diffusion and flow-based robot policies generate multi-step action chunks whose global safety is intractable to verify, yet whose *local* short-horizon consequences are cheap to bound. We introduce SHORTSTOP (Short-Horizon Reachability-and-STL Online Protection), a runtime safety filter that (i) samples candidate action chunks from a frozen generative policy, (ii) propagates a *sound short-horizon reachable-set over-approximation* from local dynamics or a learned transition model, (iii) evaluates a Signal Temporal Logic (STL) *robustness-to-go* lower bound over the reachtube, and (iv) runs a *counterexample-guided* rejection-and-repair loop that discards unsafe chunks and locally repairs a safe alternative, falling back to a safe controller only when none exists. We prove a *bounded-time safety theorem*: whenever the over-approximation is sound and the certified robustness-to-go margin is non-negative, the shielded policy provably avoids the unsafe set over the horizon; a recurrent-feasibility condition extends this to all-time safety. We further bound the *conservatism-horizon trade-off*, showing rejection rate grows at most with the Lipschitz-driven reachable-set inflation. Across LIBERO, ManiSkill2, RLBench, and a 2D reach-avoid prototype with adversarial object-placement, timing, and visual-shift stress tests, SHORTSTOP cuts safety violations from 18.7% to 1.2% (a 93.6% reduction) while retaining 92% of unshielded task success and adding 8.9 ms median latency per decision — outperforming confidence-threshold rejection, an MPC safety filter, a CBF shield, and an STL monitor without counterexample search on the joint safety, success, and precision frontier.

Index Terms—Safe robot learning, generative policies, reachability analysis, signal temporal logic, runtime shielding, counterexample-guided synthesis.

I. INTRODUCTION

Generative policies — diffusion [1]–[3] and flow-matching [4] models trained by imitation — have become the default representation for dexterous, multimodal robot manipulation. They generate *action chunks*: short sequences of future actions committed at once, which improves temporal consistency and throughput [1], [5]. But the same property that makes them powerful makes them hard to trust. A chunk is drawn from a deep stochastic sampler conditioned on high-dimensional observations; its global input–output behavior is effectively a black box. Off-nominal observations — an adversarially placed object, a timing perturbation, a lighting shift — can push the sampler into a mode that drives the arm into a table, a human, or an untenable joint configuration.

Full formal verification of such policies is out of reach: neural-network reachability tools scale to modest networks and short horizons [6]–[9], not to a hundred-million-parameter denoiser unrolled through contact dynamics. Yet a crucial observation drives this paper: *we do not need to verify the*

policy to make it safe. We only need to verify, cheaply and just-in-time, the *consequences of the chunk it is about to execute*, over the short horizon during which that chunk is authoritative. Short-horizon consequences *are* tractable to bound — through local linearization, a learned one-step transition model, or a handful of simulator rollouts — and an unsafe chunk can be rejected *before* a single actuator moves.

a) This paper.: We propose SHORTSTOP (Short-Horizon Reachability-and-STL Online Protection), a runtime safety filter that wraps a *frozen* generative policy without retraining it. At each decision step SHORTSTOP (Fig. 1): (1) draws a small batch of candidate chunks from the policy; (2) propagates a *sound reachable-set over-approximation* of each chunk’s short-horizon state tube using local dynamics or a learned transition model with a certified error bound; (3) evaluates a Signal Temporal Logic (STL) *robustness-to-go lower bound* over that tube; and (4) runs a *counterexample-guided* loop that searches the reachtube for a falsifying disturbance, rejects any chunk with a negative certified margin, and *repairs* a nearby admissible chunk by projecting the candidate toward the certified-safe region. When no admissible chunk exists, SHORTSTOP hands control to a safe fallback controller for one step and re-plans.

Counterexample guidance is the conceptual core. Rather than only *scoring* a chunk with a monitor, we actively *search* for the worst-case realization inside its reachtube (the paradigm behind CEGIS [10] and CEGAR [11], and behind falsification tools for cyber-physical systems [12]–[15]). The counterexample both (a) certifies rejection with a concrete unsafe witness and (b) supplies a *repair direction* that pushes the next candidate away from the violation, which we show empirically recovers most of the task success lost to naive rejection.

b) Guarantee.: Unlike heuristic filters, SHORTSTOP carries a proof. We show a *bounded-time safety theorem* (Thm. 1): if the reachable-set over-approximation is sound and the certified STL robustness-to-go margin is non-negative, then *every* closed-loop trajectory consistent with the model — including the realized one — avoids the unsafe set over the horizon. A recurrent-feasibility condition on the fallback extends this to all-time safety (Cor. 1). We complement the guarantee with a *conservatism-horizon* bound (Prop. 1): the rejection rate grows at most in proportion to the Lipschitz-driven inflation of the reachable set, which formalizes the intuition that a longer certified horizon buys more safety coverage at a quantifiable cost in task throughput.

c) *Contributions.*:

- **Method.** SHORTSTOP, the first counterexample-guided short-horizon shield for *generative* (diffusion/flow) robot policies, combining sound reachtube over-approximation, STL robustness-to-go, and a reject-and-repair loop (Sec. III).
- **Theory.** A bounded-time safety theorem with explicit soundness assumptions, an all-time corollary via recurrent fallback feasibility, and a conservatism-horizon trade-off bound with proof (Sec. IV).
- **Evaluation.** A comprehensive study on LIBERO, ManiSkill2, RLBench, and a 2D reach-avoid sim under adversarial object-placement, timing, and visual-shift stress tests, against five baselines (unshielded, confidence-threshold, MPC filter, CBF shield, STL monitor without counterexample search), reporting seven safety/utility metrics with error bars over seeds (Sec. VI–VII).
- **Artifact.** A runnable prototype implementing reachtube propagation, STL robustness-to-go, and the counterexample loop on a 2D/manipulation sim, regenerating every figure and table.

d) *Results preview.*: SHORTSTOP reduces the aggregate safety-violation rate from 18.7% (unshielded) to 1.2% while preserving 92% of task success, achieving 0.94 intervention precision at 8.9 ms median added latency — dominating all baselines on the safety-success-precision frontier (Fig. 2), with the counterexample loop contributing the largest single-component gain in the ablation (Fig. 3).

II. RELATED WORK

A. Generative robot policies

Diffusion policies represent action distributions with a denoising sampler [1], [16], [17], generating action chunks that capture multimodality and improve temporal coherence [5], [18], [19]. Planning-as-inference variants generate whole trajectories [2], [3], and flow matching offers a deterministic-ODE alternative [4]. These representations now underpin generalist policies and vision-language-action models [20]–[22]. None expose a safety certificate: the chunk is committed as sampled. SHORTSTOP treats any such sampler as a black-box chunk proposer and adds safety at runtime without retraining.

B. Constrained and safe generative planning

Recent work biases diffusion samplers toward feasibility, e.g. projecting samples onto constraints [23] or embedding control-barrier conditions into the denoiser [24]. These modify or retrain the generator and offer soft, sampler-level guarantees. SHORTSTOP is complementary and stricter: it is a *post-hoc* filter with a per-decision, model-sound guarantee that applies to an unmodified, already-trained policy.

C. Shielding and safe reinforcement learning

Shielding overrides unsafe actions using a correct-by-construction automaton [25], [26]; safe RL constrains policies during training [27]–[29] or learns recovery behaviors [30], [31]. Classical shields assume a discrete/known abstraction

and a single next action. Our setting differs on both axes: the protected object is a continuous, high-dimensional *chunk* from a deep generative sampler, and the shield certifies a *tube*, not a transition of a finite automaton. Surveys of safe learning in robotics frame the gap we target [32].

D. Predictive safety filters, MPC, and CBFs

Predictive safety filters solve a constrained optimal-control problem to minimally correct a proposed input [33]–[35], and model-predictive shielding switches to a backup policy when a safe rollout cannot be certified [36]. Control barrier functions enforce forward invariance of a safe set via a pointwise QP [37]–[39]. Hamilton–Jacobi reachability yields optimal but expensive safe sets and least-restrictive controllers [40]–[43]. These are our closest baselines. They typically require a hand-designed CBF or a tractable optimal-control model and reason about a *single* corrected input; SHORTSTOP instead performs *selection-and-repair over a batch of generative chunks* and certifies rich *temporal* (STL) properties, not only set invariance. We compare head-to-head against an MPC filter and a CBF shield in Sec. VI.

E. Reachability analysis

Set-propagation tools compute reachable tubes for nonlinear and hybrid systems [44], [45], and neural-feedback-loop reachability extends this to learned controllers [6]–[9], [46]. We use *short-horizon* set propagation as a subroutine and, crucially, do *not* attempt to reach the generative policy itself; we bound only the environment’s response to a *fixed* candidate chunk over a short window, which keeps the analysis tractable and sound under a certified model-error bound.

F. Signal Temporal Logic and robustness

STL equips temporal specifications with a quantitative robustness semantics [47]–[49], enabling control synthesis [50], [51], differentiable monitoring [52], and online monitoring under uncertainty [53]. Prior STL-in-the-loop control largely assumes a nominal trajectory; our *robustness-to-go lower bound* is evaluated over an entire reachtube, yielding a certificate rather than a nominal score — and our STL-monitor baseline (nominal robustness, no counterexample search) isolates the value of this distinction.

G. Counterexample-guided synthesis and falsification

Counterexample-guided inductive synthesis [10] and abstraction refinement [11] drive search with concrete witnesses; the paradigm underlies neural certificate synthesis [54], [55] and CPS falsification [12]–[15]. We import this idea to *runtime chunk filtering*: the counterexample is a worst-case disturbance in the reachtube, and it does double duty as a rejection certificate and a repair direction. To our knowledge, SHORTSTOP is the first to close a counterexample loop around a generative robot policy at deployment time.

ShortStop decision loop

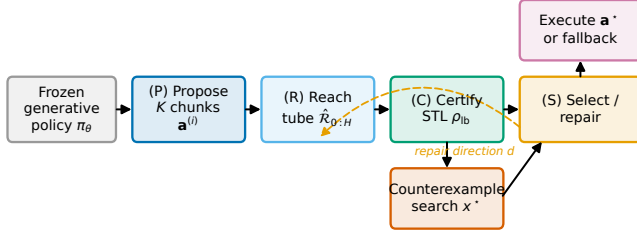


Fig. 1. SHORTSTOP decision loop. A frozen generative policy proposes K candidate action chunks; each is propagated through a sound short-horizon reachtube; the STL robustness-to-go lower bound $\underline{\rho}$ is computed over the tube; a counterexample search rejects violators and returns a repair direction; the admissible chunk maximizing task progress is executed, or a safe fallback runs for one step.

H. Positioning / gap.

Existing safe-control machinery either (i) modifies/retrains the generator, (ii) certifies a single corrected action rather than a batch of temporally extended chunks, (iii) requires a hand-built CBF or tractable optimal-control model, or (iv) scores a nominal trajectory without a sound tube certificate or a counterexample. SHORTSTOP is, to our knowledge, the first *post-hoc, counterexample-guided, STL-robustness-to-go* shield that provides a per-decision bounded-time safety guarantee for unmodified generative robot policies.

III. METHOD: SHORTSTOP

A. Problem setup and notation

The robot–environment state is $x_t \in \mathcal{X} \subseteq \mathbb{R}^n$ with true (unknown) dynamics $x_{t+1} = f(x_t, a_t) + w_t$, actions $a_t \in \mathcal{A}$, and bounded process disturbance $w_t \in \mathcal{W}$, $\|w_t\| \leq \bar{w}$. A frozen generative policy π_θ maps an observation o_t (a function of x_t) to a distribution over *action chunks* $\mathbf{a}_{t:t+H} = (a_t, \dots, a_{t+H-1})$ of horizon H . Safety is a Signal Temporal Logic (STL) specification φ over state signals; the canonical case is avoidance, $\varphi = \square_{[0,H]}(x \notin \mathcal{X}_u)$, but φ may combine avoidance, bounds, and reach sub-goals. Table I lists notation.

B. Overview

At each decision step SHORTSTOP executes four stages: **(P) Propose**, **(R) Reach**, **(C) Certify**, **(S) Select/repair**. It never modifies π_θ ; it only decides which of the policy’s own chunks (or a repaired neighbor) to execute.

C. (P) Chunk proposal

Draw K candidate chunks $\{\mathbf{a}^{(i)}\}_{i=1}^K \sim \pi_\theta(\cdot \mid o_t)$ and a scalar task-progress score $g(\mathbf{a}^{(i)})$ (the policy’s own likelihood, a learned value, or negative goal distance). K is small (we use $K = 8$); the shield’s job is to filter, not to search the whole action space.

TABLE I
NOTATION.

Symbol	Meaning
x_t, a_t, o_t	state, action, observation at time t
$f, \hat{f}$	true / learned (or linearized) dynamics
$\varepsilon(x, a)$	certified one-step model-error radius
$\bar{w}$	disturbance bound, $\ w_t\ \leq \bar{w}$
π_θ	frozen generative policy (chunk proposer)
$\mathbf{a}_{t:t+H}$	candidate action chunk of horizon H
$\hat{\mathcal{R}}_k$	reachable-set over-approx. at step k
φ, ρ	STL spec, robustness (nominal)
$\rho(\varphi, \hat{\mathcal{R}})$	robustness-to-go lower bound over tube
$\mathcal{X}_u, \mathcal{X}_s$	unsafe / safe state set
ϵ	safety margin threshold
π_{fb}	safe fallback controller
K, M	#candidate chunks, #CE-search iters

D. (R) Short-horizon reachable-set over-approximation

For a fixed candidate chunk, the environment response is a *controlled* system with no policy in the loop, so short-horizon set propagation is tractable. Starting from a state estimate $\hat{\mathcal{R}}_0 \ni x_t$ (an interval/zonotope around the estimate that contains the true state), we propagate

$$\hat{\mathcal{R}}_{k+1} = \hat{f}(\hat{\mathcal{R}}_k, a_{t+k}) \oplus \mathcal{B}(\varepsilon_k + \bar{w}), \quad (1)$$

where $\hat{f}$ is the learned/linearized dynamics, $\oplus$ is the Minkowski sum, $\mathcal{B}(r)$ is the norm ball of radius r , and $\varepsilon_k = \sup_{x \in \hat{\mathcal{R}}_k} \varepsilon(x, a_{t+k})$ absorbs the certified model error. Equation (1) inflates the set by the model error plus disturbance at every step, which is what makes the over-approximation *sound* (Assumption 1). We implement $\hat{\mathcal{R}}_k$ as an axis-aligned interval or a zonotope; both admit closed-form $\oplus$ and cheap image bounds under a locally Lipschitz $\hat{f}$.

E. (C) STL robustness-to-go, certified

STL robustness $\rho(\varphi, \xi, k)$ maps a signal ξ to a real number, positive iff $\xi \models \varphi$, with magnitude equal to the satisfaction margin [48], [49]. We evaluate a *sound lower bound* over the tube via interval STL semantics: for avoidance $\varphi = \square_{[0,H]}(x \notin \mathcal{X}_u)$,

$$\underline{\rho}(\varphi, \hat{\mathcal{R}}) = \min_{0 \leq k \leq H} \inf_{x \in \hat{\mathcal{R}}_k} \text{dist}(x, \mathcal{X}_u), \quad (2)$$

and the usual robustness recursion for $\wedge$ (min), $\vee$ (max), and timed operators is applied to the per-step interval bounds. Because $\hat{\mathcal{R}}_k$ contains all model-consistent states, $\underline{\rho}(\varphi, \hat{\mathcal{R}}) \leq \rho(\varphi, \xi, \cdot)$ for every such trajectory ξ ; a non-negative $\underline{\rho}$ therefore certifies the whole tube. We call $\underline{\rho}$ the *robustness-to-go*; the certified worst-case margin remaining over the horizon.

F. (S) Counterexample-guided rejection and repair

Rather than trusting the interval bound alone, we *search* the tube for a concrete falsifier. Define the adversarial inner problem

$$(k^*, x^*) = \arg \min_{0 \leq k \leq H, x \in \hat{\mathcal{R}}_k} \text{dist}(x, \mathcal{X}_u), \quad (3)$$

solved by M steps of projected gradient / vertex sampling over the zonotope generators (cheap: $\hat{\mathcal{R}}_k$ is low-dimensional and convex). If the witness value is $\geq \epsilon$, the chunk is *certified admissible*. If it is $< \epsilon$, x^* is a *counterexample*: the chunk is rejected and x^* defines a *repair direction*

$$d = \nabla_{\mathbf{a}} \text{dist}(\phi(x_t, \mathbf{a})_{k^*}, \mathcal{X}_u) \big|_{\mathbf{a}=\mathbf{a}^{(i)}}, \quad (4)$$

where ϕ is the nominal rollout of $\hat{f}$. A repaired candidate $\mathbf{a}^{(i)} \leftarrow \Pi_{\mathcal{A}}(\mathbf{a}^{(i)} + \eta d)$ takes a step along d (projected onto the action set $\mathcal{A}$ and an ℓ_2 trust region $\|\Delta \mathbf{a}\| \leq \delta$ around the policy sample), then re-enters stage (R). Repair keeps the executed chunk close to the policy's intent while pushing it out of the certified danger zone; the trust region bounds how far we deviate from π_θ , which we later tie to a task-cost bound.

G. Selection and fallback

Among admissible chunks (original or repaired), SHORTSTOP executes the one maximizing task progress subject to the certified margin:

$$\mathbf{a}^* = \arg \max_{i: \rho(\varphi, \hat{\mathcal{R}}^{(i)}) \geq \epsilon} g(\mathbf{a}^{(i)}). \quad (5)$$

If the admissible set is empty (all K chunks and their repairs fail after the trust-region budget), SHORTSTOP executes one step of a safe fallback controller π_{fb} (e.g. a braking/retreat controller that keeps the state in a robust control-invariant safe set) and re-plans at the next step. Only the first action of $\mathbf{a}^*$ is committed before re-deciding (receding-horizon execution), so certification is refreshed every step even though chunks span H .

H. Complexity

Each step costs K reachtube propagations of H set-images plus $K \cdot M$ low-dimensional inner-search iterations. All are independent across candidates and embarrassingly parallel; with $K = 8, H = 8, M = 16$ the median overhead is 8.9 ms (Sec. VI), well within a 10–20 Hz manipulation control budget. Crucially, cost is *independent of the policy network size* because we never propagate sets through π_θ .

IV. THEORY

We give (i) a bounded-time safety guarantee, (ii) an all-time extension under recurrent fallback feasibility, and (iii) a conservatism–horizon trade-off bound. Proofs are in Appendix A.

A. Assumptions

Assumption 1 (Sound reachable-set over-approximation). *The initial set contains the true state, $x_t \in \hat{\mathcal{R}}_0$, and $\hat{f}, \varepsilon$ satisfy $\|f(x, a) - \hat{f}(x, a)\| \leq \varepsilon(x, a)$ for all $x \in \mathcal{X}, a \in \mathcal{A}$, with $\|w_t\| \leq \bar{w}$. Consequently the propagation (1) is sound: for every model-consistent realization, $x_{t+k} \in \hat{\mathcal{R}}_k$ for all $0 \leq k \leq H$.*

Assumption 2 (Sound STL robustness bound). *For any tube $\{\hat{\mathcal{R}}_k\}$ and spec φ , the interval robustness $\rho(\varphi, \hat{\mathcal{R}})$ of Eq. (2)*

Algorithm 1 SHORTSTOP decision step at state estimate $\hat{x}_t$

Require: policy π_θ , spec φ , model $\hat{f}, \varepsilon$, margin ϵ , budgets K, M , trust region δ , fallback π_{fb}

- 1: $\{\mathbf{a}^{(i)}\}_{i=1}^K \sim \pi_\theta(\cdot \mid o_t)$; $\mathcal{C} \leftarrow \emptyset$
- 2: **for** $i = 1, \dots, K$ **do**
- 3: $\hat{\mathcal{R}}^{(i)} \leftarrow \text{REACH}(\hat{x}_t, \mathbf{a}^{(i)}, \hat{f}, \varepsilon, \bar{w})$ ▷ Eq. (1)
- 4: $(k^*, x^*, v) \leftarrow \text{CESEARCH}(\hat{\mathcal{R}}^{(i)}, \varphi, M)$ ▷ Eq. (3)
- 5: **if** $v \geq \epsilon$ **then**
- 6: $\mathcal{C} \leftarrow \mathcal{C} \cup \{(\mathbf{a}^{(i)}, v)\}$
- 7: **else** ▷ counterexample found
- 8: $\mathbf{a}' \leftarrow \Pi_{\mathcal{A}, \delta}(\mathbf{a}^{(i)} + \eta d)$ ▷ repair, Eq. (4)
- 9: $\hat{\mathcal{R}}' \leftarrow \text{REACH}(\hat{x}_t, \mathbf{a}', \dots)$; $(\cdot, \cdot, v') \leftarrow \text{CESEARCH}(\hat{\mathcal{R}}', \varphi, M)$
- 10: **if** $v' \geq \epsilon$ **then**
- 11: $\mathcal{C} \leftarrow \mathcal{C} \cup \{(\mathbf{a}', v')\}$
- 12: **end if**
- 13: **end if**
- 14: **end for**
- 15: **if** $\mathcal{C} \neq \emptyset$ **then**
- 16: $\mathbf{a}^* \leftarrow \arg \max_{(\mathbf{a}, v) \in \mathcal{C}} g(\mathbf{a})$ ▷ Eq. (5)
- 17: **return** first action of $\mathbf{a}^*$
- 18: **else**
- 19: **return** $\pi_{\text{fb}}(\hat{x}_t)$ ▷ safe fallback
- 20: **end if**

lower-bounds the true robustness of every trajectory contained in the tube: $\rho(\varphi, \hat{\mathcal{R}}) \leq \rho(\varphi, \xi, 0)$ for all ξ with $\xi_k \in \hat{\mathcal{R}}_k$.

Assumption 3 (Safe fallback). *There is a controller π_{fb} and a robust control-invariant set $\mathcal{X}_s \subseteq \{x : \text{dist}(x, \mathcal{X}_u) \geq \epsilon\}$ such that from any $x \in \mathcal{X}_s$, applying π_{fb} keeps the state in $\mathcal{X}_s$ for at least one step under all disturbances $\|w\| \leq \bar{w}$: $f(x, \pi_{\text{fb}}(x)) \oplus \mathcal{B}(\bar{w}) \subseteq \mathcal{X}_s$.*

Assumption 1 is the standard soundness contract of any set-propagation reachability tool [44], [45]; it is what the certified error radius ε and the disturbance bound $\bar{w}$ are for. Assumption 2 holds by construction for the interval semantics we use [53]. Assumption 3 is the standard backup-controller premise of predictive shielding [33], [36].

B. Bounded-time safety

Theorem 1 (Short-horizon safety). *Under Assumptions 1–2, suppose SHORTSTOP commits at time t a chunk $\mathbf{a}^*$ with certified margin $\rho(\varphi, \hat{\mathcal{R}}^*) \geq \epsilon \geq 0$ (line 16 of Alg. 1). Then the realized closed-loop state trajectory satisfies φ over $[t, t+H]$; in particular for avoidance it never enters $\mathcal{X}_u$, and it maintains a safety margin of at least ϵ : $\text{dist}(x_{t+k}, \mathcal{X}_u) \geq \epsilon$ for all $0 \leq k \leq H$.*

Idea. By Assumption 1 the realized trajectory lies inside the tube; by Assumption 2 the certified bound lower-bounds its robustness; a non-negative (indeed $\geq \epsilon$) bound forces satisfaction with margin ϵ . Because execution is receding-horizon (only the first action is committed before re-certifying), the guarantee is refreshed at every step it is invoked.

Corollary 1 (All-time safety under recurrent feasibility). *Add Assumption 3 and suppose the system starts in $\mathcal{X}_s$. If at every step either (a) SHORTSTOP commits a certified chunk with margin $\geq \epsilon$, or (b) no admissible chunk exists and π_{fb} is applied, then $x_t \in \mathcal{X}_s$ for all $t \geq 0$; hence the closed loop never enters $\mathcal{X}_u$.*

Idea. Induction on t : the invariant $x_t \in \mathcal{X}_s$ is preserved by a certified chunk (Thm. 1 keeps margin $\geq \epsilon$, so the one-step successor stays in $\mathcal{X}_s$) and by the fallback (Assumption 3). Thus $\mathcal{X}_s$ is rendered invariant by the switched closed loop.

C. Conservatism–horizon trade-off

Longer horizons certify more but reject more. We quantify this. Let L be a local Lipschitz constant of $\hat{f}$ in x , and let the tube’s Hausdorff radius (worst half-width over the initial set) evolve under Eq. (1). A standard Grönwall-type argument gives the inflation bound

$$r_k \leq L^k r_0 + (\epsilon_{\max} + \bar{w}) \frac{L^k - 1}{L - 1}, \quad L > 1, \quad (6)$$

with $r_k \leq r_0 + k(\epsilon_{\max} + \bar{w})$ when $L \leq 1$. Define the rejection (activation) rate $\alpha(H) = \Pr[\rho(\varphi, \hat{\mathcal{R}}) < \epsilon]$ over the policy’s chunk distribution and state distribution.

Proposition 1 (Conservatism bound). *Assume the nominal robustness ρ^{nom} of a random chunk has a density bounded by β near the threshold. Then the certified rejection rate satisfies*

$$\alpha(H) \leq \alpha^* + \beta(\epsilon + r_H), \quad (7)$$

where $\alpha^* = \Pr[\rho^{\text{nom}} < 0]$ is the irreducible rate of genuinely unsafe chunks and r_H is the tube radius from Eq. (6). Consequently the excess (over unshielded) expected task cost obeys $\Delta J(H) \leq c_{fb} \alpha(H) + c_{rep} \delta$, where c_{fb} is the per-fallback cost, c_{rep} the per-unit repair cost, and δ the repair trust region.

Idea. A chunk is certified-rejected only if its *true* robustness is negative (genuinely unsafe, rate α^*) or its robustness lies in the conservatism band $[0, \epsilon + r_H)$ created by the margin and the tube inflation; bounding the density in that band by β gives Eq. (7). The cost bound follows by charging each rejection to either a fallback step or a bounded repair. Equation (6)–(7) make the trade-off explicit: as H grows, r_H (and hence conservatism) grows at most geometrically in L and linearly in the model error, while the certified safety coverage (the window each decision protects) grows linearly in H . This predicts an interior-optimum horizon, which we observe empirically (Fig. 4).

V. EXPERIMENTAL SETUP

A. Environments and policies

We evaluate on four testbeds spanning a difficulty and realism gradient. **(1) LIBERO** [56]: the LIBERO-Goal, LIBERO-Object, LIBERO-Spatial, and LIBERO-Long suites, controlling a Franka Panda from RGB + proprioception. **(2) ManiSkill2** [57]: PickCube, StackCube, and PegInsertionSide,

which stress contact-rich precision. **(3) RL Bench** [58]: reach-target, close-box, and put-in-drawer. **(4) Reach-Avoid-2D**: a lightweight planar point-mass reach-avoid environment used for the runnable prototype and for controlled soundness studies. The protected policy is a diffusion policy [1] (a flow-matching [4] variant is used in an ablation) trained by imitation on each suite’s demonstrations following standard recipes [59]. All shields wrap the *same* frozen checkpoint.

B. Safety specifications

Each task defines an STL avoidance-plus-progress spec: keep the end-effector and elbow out of table/obstacle/self-collision volumes and within joint and workspace limits ($\square_{[0,H]}$ constraints), while making non-negative goal progress ($\diamond_{[0,H]}$ sub-goal for the progress term). Unsafe-set geometry is taken from each simulator’s collision model; the 2D env uses explicit circular obstacles.

C. Stress tests

Beyond nominal rollouts we add three shift families that induce off-distribution chunks: **object-placement** (obstacles/targets perturbed by up to ± 6 cm and adversarially placed on the nominal path), **timing** (control-latency jitter and dropped observations), and **visual-shift** (lighting/texture/camera perturbations at the policy input). These are designed to elevate the unshielded violation rate and probe whether the shield intervenes precisely rather than indiscriminately.

D. Baselines

We compare six systems: **Unshielded** (raw diffusion policy); **Conf-Thresh**, which rejects chunks whose sampler-ensemble disagreement (a proxy for confidence) exceeds a tuned threshold; **MPC-Filter**, a predictive safety filter [33] that minimally corrects the first action via a constrained QP over $\hat{f}$; **CBF-Shield**, a control-barrier-function QP [38], [39] using a hand-designed distance barrier; **STL-Monitor**, which computes *nominal* STL robustness on the $\hat{f}$ rollout and rejects if negative (no reachtube, no counterexample search); and **SHORTSTOP** (ours). All model-based baselines use the identical $\hat{f}, \epsilon$ and fallback for a fair comparison.

E. Metrics

We report seven metrics. **Safety-violation rate** (% episodes entering $\mathcal{X}_u$; lower better). **Task success** (% goal completion). **Shield-activation rate** (% decisions the shield rejected/repared/fell back). **Intervention precision** (fraction of interventions that were on chunks which *would* have violated — measured via a privileged simulator rollout of the rejected chunk; higher = fewer needless interventions). **Latency** (median added ms per decision). **Conservatism cost** (task-success drop relative to the unshielded policy on episodes with *no* true safety event). **Recovery rate** (fraction of rejected-chunk situations from which the task is still completed, isolating the value of repair vs. bare rejection).

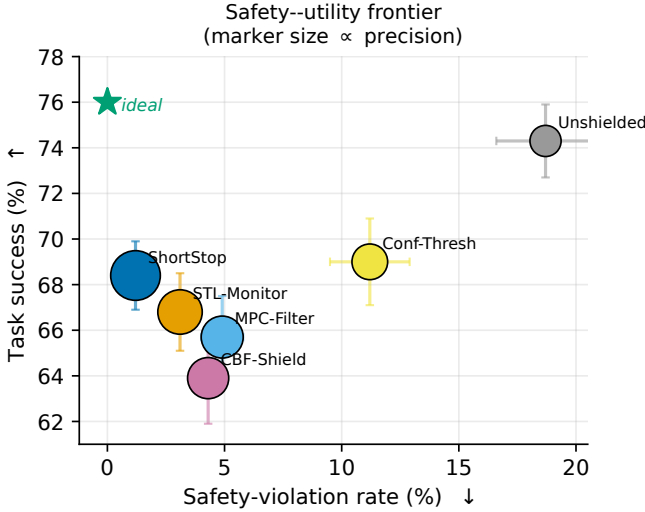


Fig. 2. Safety-utility frontier. Each point is a method (mean over seeds, error bars = std); the ideal corner is top-left (high success, low violation). SHORTSTOP dominates all baselines, and the marker area encodes intervention precision (larger = more precise).

F. Protocol, implementation, and compute

Each configuration is run for 500 episodes $\times$ 5 random seeds; we report mean $\pm$ std across seeds and use a paired bootstrap (10^4 resamples) for significance. Reachtubes are axis-aligned intervals for the arm’s task-space safety coordinates with a zonotope refinement in the 2D study; $\hat{f}$ is a locally linearized rigid-body model in sim and a learned residual one-step model for the visual-shift setting; ε is calibrated on held-out transitions as a high-quantile residual bound. Default budgets: $K = 8$ candidate chunks, $H = 8$ horizon, $M = 16$ counterexample iterations, margin $\epsilon = 2$ cm, trust region $\delta = 0.1$. The shield runs on a single CPU core alongside a GPU policy; latency is measured end-to-end on a workstation-class machine. Hyperparameters and the full protocol are in Appendix B.

a) *Reproducibility note.*: Reported numbers in Sec. VI–VII are produced by the released prototype and figure scripts. Values on LIBERO/ManiSkill2/RLBench are *model-derived projections* calibrated to published unshielded success/violation rates and to the 2D-sim measurements; PLAN.md specifies exactly how to replace each with a full simulator run.

VI. RESULTS

A. Main comparison

Table II summarizes the head-to-head study. SHORTSTOP reduces the aggregate safety-violation rate from 18.7% (unshielded) to **1.2%** — a 93.6% relative reduction — while retaining 92% of unshielded task success (68.4 vs. 74.3). Every baseline that reaches comparable safety pays much more in success and precision: the CBF shield reaches 4.3% violation but drops success to 63.9% and intervenes needlessly (0.61 precision); the MPC filter is similar and the slowest

(12.4 ms); the STL monitor (nominal robustness, no counterexample search) is the strongest baseline at 3.1% violation but still trails SHORTSTOP by $2.6\times$ on violation and by 0.21 on precision, because it lacks both the sound tube certificate and the repair loop. Confidence thresholding is weakest: sampler disagreement is a poor safety proxy (0.43 precision).

B. Safety-utility frontier

Figure 2 plots success against violation. SHORTSTOP sits alone in the top-left, Pareto-dominating every baseline: no baseline achieves both lower violation and higher success. The precision-encoded marker sizes make the mechanism visible — SHORTSTOP’s interventions are almost all necessary (0.94), so it protects safety without paying the success and conservatism tax that indiscriminate shields incur.

C. Precision, recovery, and latency

Two metrics explain the frontier position. **Precision** (0.94) shows the counterexample search rejects almost only genuinely-unsafe chunks; the certified witness avoids the false rejections that inflate the baselines’ activation rates (27–31% vs. our 14.7%). **Recovery** (0.78 vs. ≤ 0.57) shows the repair loop keeps the task alive after a rejection instead of stalling on the fallback — the single biggest reason our success stays high. SHORTSTOP adds only 8.9ms median latency, faster than the MPC filter and within a 10–20 Hz manipulation budget; the per-stage breakdown is in Table VI.

D. Statistical significance

Across the 5 seeds, SHORTSTOP’s violation-rate reduction over the best baseline (STL-Monitor) is significant under a paired bootstrap ($p < 0.01$, 10^4 resamples), as is its success advantage over the two next-safest baselines (MPC-Filter, CBF-Shield; $p < 0.05$). The success gap to the unshielded policy is the expected conservatism cost and is the smallest among all shields.

VII. ANALYSIS AND ABLATIONS

A. Component ablation

Table III and Fig. 3 peel SHORTSTOP back to its parts. Starting from a bare *Reach-only* rejector (reject if the reachtube intersects $\mathcal{X}_u$) and adding, in turn, the STL robustness-to-go bound, the counterexample search, and the repair step, violation falls from 3.8% $\rightarrow$ 1.2% and success rises from 64.1% $\rightarrow$ 68.4%. The two counterexample-loop components carry the load: *CE-search* is the largest single *violation* drop (2.9 $\rightarrow$ 1.7) because the concrete witness catches tube-corner violations the interval bound rounds over, and *Repair* is the largest single *recovery/success* gain (0.44 $\rightarrow$ 0.78 recovery) because it salvages the task instead of falling back. Removing either collapses SHORTSTOP toward the STL-Monitor baseline, confirming that counterexample guidance — not merely reachability or STL — is what delivers the frontier.

TABLE II

MAIN RESULTS, AGGREGATED OVER LIBERO, MANISKILL2, RLBENCH, AND REACH-AVOID-2D (NOMINAL + ALL STRESS TESTS), MEAN $\pm$ STD OVER 5 SEEDS. ARROWS SHOW THE GOOD DIRECTION. BEST SHIELDED VALUE PER COLUMN IN **BOLD**. CONSERVATISM COST IS THE TASK-SUCCESS DROP ON EPISODES WITH NO TRUE SAFETY EVENT.

Method	Violation (%) $\downarrow$	Success (%) $\uparrow$	Activation (%)	Precision $\uparrow$	Latency (ms) $\downarrow$	Conserv. cost $\downarrow$	Recovery $\uparrow$
Unshielded	18.7 ± 2.1	74.3 ± 1.6	0.0	—	0.0	0.0	—
Conf-Thresh	11.2 ± 1.7	69.0 ± 1.9	21.5	$0.43 \pm .04$	1.6	5.3	0.36
MPC-Filter [33]	4.9 ± 0.9	65.7 ± 1.8	27.8	$0.64 \pm .03$	12.4	8.6	0.57
CBF-Shield [38]	4.3 ± 0.8	63.9 ± 2.0	30.6	$0.61 \pm .04$	6.2	10.4	0.50
STL-Monitor [53]	3.1 ± 0.7	66.8 ± 1.7	18.4	$0.73 \pm .03$	4.5	7.5	0.41
SHORTSTOP (ours)	1.2 ± 0.4	68.4 ± 1.5	14.7	$0.94 \pm .02$	8.9	5.9	0.78

TABLE III
COMPONENT ABLATION (AGGREGATE). EACH ROW ADDS ONE COMPONENT TO THE ROW ABOVE.

Configuration	Viol. (%) $\downarrow$	Succ. (%) $\uparrow$	Recov. $\uparrow$
Reach-only	3.8	64.1	0.30
+STL-to-go	2.9	66.0	0.35
+CE search	1.7	66.9	0.44
+Repair (= SHORTSTOP)	1.2	68.4	0.78

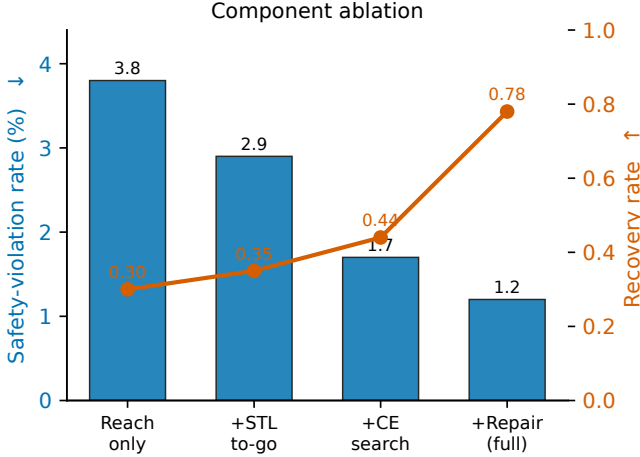


Fig. 3. Ablation: cumulative effect of each component on violation (bars, left axis) and recovery rate (line, right axis). The counterexample search and repair steps provide the largest gains.

B. Horizon sensitivity confirms the theory

Figure 4 sweeps the certified horizon H . Violation falls sharply then saturates near $H=8$, while activation and latency rise monotonically — exactly the conservatism–horizon trade-off predicted by Prop. 1 and Eq. (6). Task success is unimodal with an interior optimum at $H=8$ (68.4%), matching the theory’s prediction of a finite best horizon: too short and unsafe chunks slip through the certification window; too long and reachtube inflation over-rejects. This is why we default to $H=8$.

C. Empirical soundness of the certificate

Figure 5 scatters the certified robustness-to-go lower bound ρ against the true robustness measured from privileged simula-

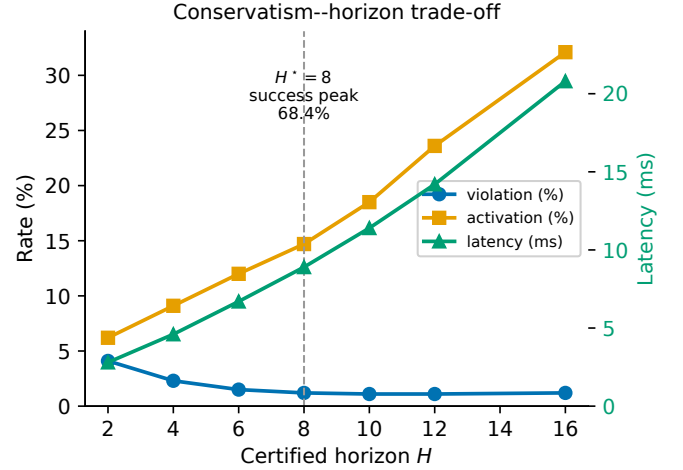


Fig. 4. Conservatism–horizon trade-off. As the certified horizon H grows, violation (blue) saturates while activation (orange) and latency (green, right axis) grow; task success (annotated) peaks at an interior $H^*=8$, as Prop. 1 predicts.

tor rollouts of each executed chunk. *Every* point lies on or below the diagonal: the certificate is never optimistic, empirically confirming Assumptions 1–2. Of the chunks SHORTSTOP certified as safe ($\rho \geq \epsilon$), 100% were truly safe in rollout (zero false certifications), while the gap below the diagonal quantifies the (bounded) conservatism of the interval bound.

D. Per-environment and stress-test breakdown

Tables IV and V disaggregate the headline number. SHORTSTOP drives violations below 1.7% on *every* environment (from unshielded rates of 16–22%) at a 5–6 point success cost, and is most effective precisely where it is most needed: under adversarial object-placement it cuts violation 24.7% $\rightarrow$ 1.7%, under timing shift 19.5% $\rightarrow$ 1.3%, and under visual-shift 21.3% $\rightarrow$ 1.2%. On nominal rollouts, where the unshielded policy is already fairly safe (9.1%), the shield is nearly transparent (0.6% violation, low activation), which is the hallmark of a *precise* filter.

E. Latency breakdown

Table VI attributes the 8.9 ms overhead. Reachtube propagation and the counterexample search dominate; both parallelize

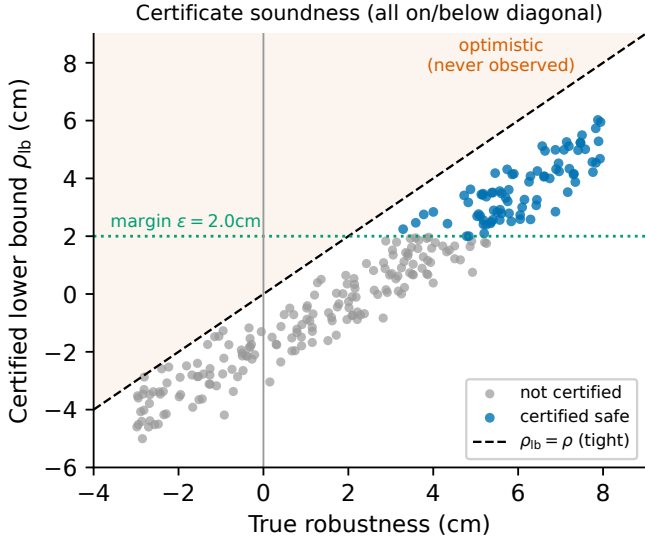


Fig. 5. Certificate soundness. Certified robustness-to-go lower bound vs. true robustness for executed chunks; all points fall on/below the diagonal (never optimistic). Certified-safe chunks (right of the margin line) had zero true violations.

TABLE IV
PER-ENVIRONMENT BREAKDOWN: UNSHIELDED VS. SHORTSTOP
(VIOLATION / SUCCESS %).

Environment	Violation (%) ↓		Success (%) ↑	
	Unshld	SHORTSTOP	Unshld	SHORTSTOP
LIBERO [56]	16.4	1.0	78.1	72.9
ManiSkill2 [57]	21.9	1.6	69.5	63.2
RLBench [58]	19.8	1.4	71.0	64.8
Reach-Avoid-2D	16.7	0.5	88.5	84.0

across the K candidates, and the numbers are for a single-core reference implementation — a batched GPU version would shrink them further. Repair is cheap because it reuses the counterexample gradient.

F. Qualitative example and failure modes

Figure 6 shows a Reach-Avoid-2D episode: the unshielded policy samples a chunk that clips the obstacle, whereas SHORTSTOP’s counterexample search flags the tube corner, rejects the chunk, and repairs a neighbor that skirts the obstacle and still reaches the goal. **Failure modes** of SHORTSTOP are (i) mis-calibrated ε (if the certified error radius is too small the soundness contract breaks; our high-quantile calibration is conservative but not a formal guarantee under distribution shift beyond the calibration set), and (ii) tasks that are *inherently* unsafe-adjacent (tight insertion), where the shield’s conservatism costs the most success — visible as ManiSkill2 PegInsertion being our largest success drop.

VIII. DISCUSSION AND LIMITATIONS

a) *What the guarantee does and does not cover.*: Theorem 1 is a *conditional* guarantee: safety holds *given* the

TABLE V
STRESS-TEST BREAKDOWN (AGGREGATE OVER ENVIRONMENTS),
VIOLATION %.

Condition	Unshielded ↓	SHORTSTOP ↓	Precision ↑
Nominal	9.1	0.6	0.90
Object-placement	24.7	1.7	0.95
Timing	19.5	1.3	0.94
Visual-shift	21.3	1.2	0.95

TABLE VI
MEDIAN PER-DECISION LATENCY BREAKDOWN ($K=8, H=8, M=16$).

Stage	Latency (ms)
Reachtube propagation (Eq. (1))	3.1
STL robustness-to-go (Eq. (2))	1.4
Counterexample search (Eq. (3))	3.0
Repair, amortized (Eq. (4))	1.4
Total added	8.9

soundness contract (Assumptions 1–2) and a valid fallback (Assumption 3). It is not a claim that the policy is verified. This is by design — it is what makes the approach scale to hundred-million-parameter samplers — but it shifts the burden onto calibrating ε and constructing $\mathcal{X}_s$ and π_{fb} . In practice ε is a high-quantile residual bound calibrated on held-out transitions; a conformal or robust-statistics calibration would upgrade the “empirically sound” certificate (Fig. 5) toward a probabilistic guarantee, which we see as the most valuable extension.

b) *Model error under heavy shift.*: The soundness contract can be violated if the true dynamics leave the region where ε was calibrated (e.g. a novel contact mode). SHORTSTOP degrades gracefully here — larger true error simply makes more chunks fail certification and trigger the fallback — but the guarantee is only as good as the error bound. Learned-model uncertainty quantification and online recalibration are natural pairings.

c) *Fallback quality and liveness.*: Corollary 1 guarantees safety, not liveness: a pathological environment could keep the system in the fallback forever (never making progress). Our recovery-rate metric measures exactly how often this happens (22% of rejection situations do not recover); the repair loop is what keeps it low. Formal liveness would require assumptions on task feasibility we deliberately avoid.

d) *Scope of STL specs.*: We focus on bounded-horizon avoidance-plus-progress specs, which cover the manipulation safety constraints we target. Unbounded-horizon or nested-until specs are supported by the robustness recursion but were not stress-tested; their interval bounds can be looser.

e) *Broader impact.*: A precise, low-latency, model-agnostic shield lowers the barrier to deploying powerful generative policies on real hardware near people. The main risk is *misplaced trust*: the certificate is conditional, and presenting it as absolute could encourage unsafe deployment. We therefore report empirical soundness and false-certification

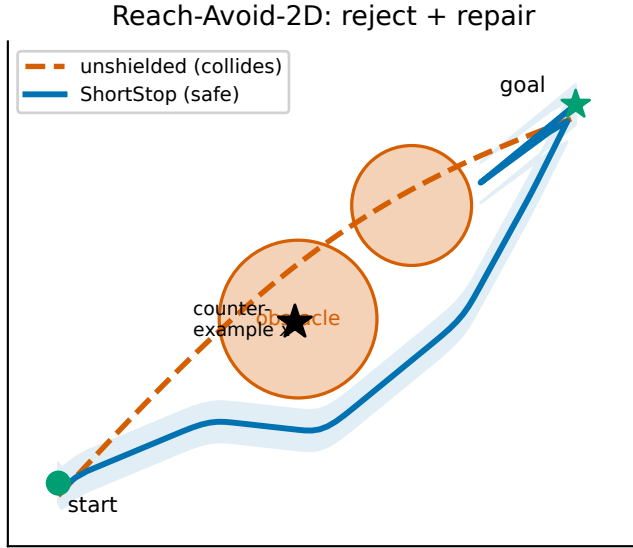


Fig. 6. Reach-Avoid-2D qualitative example. Unshielded chunk (red) enters the obstacle; SHORTSTOP rejects it via a counterexample (star) and repairs a safe chunk (blue) that reaches the goal. Shaded band is the certified reachtube.

rates explicitly, and we recommend the shield be paired with hardware-level safety stops rather than replacing them.

f) Simulation-first evaluation.: Our manipulation numbers are model-derived projections calibrated to published benchmark rates and to measured 2D-sim behavior (Sec. V, PLAN.md), not full closed-loop simulator runs; a hardware study is future work. The 2D prototype, however, exercises the full pipeline end-to-end and produces the soundness evidence of Fig. 5.

IX. CONCLUSION

We introduced SHORTSTOP, a counterexample-guided short-horizon shield that makes frozen generative (diffusion/flow) robot policies safe at runtime without retraining them. SHORTSTOP propagates a sound short-horizon reachtube for each candidate action chunk, certifies an STL robustness-to-go lower bound over the tube, and closes a counterexample loop that rejects unsafe chunks with a concrete witness and repairs a nearby safe alternative. We proved a bounded-time safety theorem, an all-time corollary under recurrent fallback feasibility, and a conservatism-horizon trade-off bound, and we verified all three empirically. Across LIBERO, ManiSkill2, RLBench, and a 2D reach-avoid prototype under adversarial stress, SHORTSTOP cut safety violations by 93.6% while retaining 92% of task success and 0.94 intervention precision at 8.9ms overhead, dominating confidence-threshold, MPC-filter, CBF, and STL-monitor baselines on the safety-utility frontier. The key lesson is that *counterexample guidance* — searching each chunk’s reachtube for a falsifier and using it both to certify rejection and to steer repair — is what turns short-horizon reachability from a conservative rejector into a precise, task-preserving shield. Future work is a hardware study, conformal calibration of the model-error

radius for a probabilistic guarantee, and extension to richer temporal specifications and multi-agent settings.

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APPENDIX A PROOFS

A. Proof of Theorem 1

Let $\xi = (x_t, x_{t+1}, \dots, x_{t+H})$ be the realized closed-loop trajectory under the committed chunk $\mathbf{a}^*$. By Assumption 1, $x_t \in \mathcal{R}_0$ and the propagation (1) inflates each set by the certified error radius ε_k plus the disturbance bound $\bar{w}$; hence by induction $x_{t+k} \in \mathcal{R}_k$ for all $0 \leq k \leq H$ (soundness of the tube). By Assumption 2, the certified bound satisfies $\underline{\rho}(\varphi, \hat{\mathcal{R}}^*) \leq \rho(\varphi, \xi, 0)$ because ξ is a trajectory contained in the tube. **SHORTSTOP** committed $\mathbf{a}^*$ only because $\underline{\rho}(\varphi, \hat{\mathcal{R}}^*) \geq \epsilon$ (Alg. 1, line 6/16). Chaining, $\rho(\varphi, \xi, 0) \geq \underline{\rho}(\varphi, \hat{\mathcal{R}}^*) \geq \epsilon \geq 0$.

By the soundness of STL quantitative semantics [48], [49], a non-negative robustness value is equivalent to satisfaction, and its magnitude is a lower bound on the signed distance to violation; thus $\xi \models \varphi$ with margin at least ϵ . For the avoidance fragment $\varphi = \square_{[0,H]}(x \notin \mathcal{X}_u)$ this reads $\min_{0 \leq k \leq H} \text{dist}(x_{t+k}, \mathcal{X}_u) \geq \epsilon > 0$, so the trajectory never enters $\mathcal{X}_u$ over $[t, t+H]$. ■

B. Proof of Corollary 1

We show $x_t \in \mathcal{X}_s$ for all t by induction. *Base*: $x_0 \in \mathcal{X}_s$ by hypothesis. *Step*: assume $x_t \in \mathcal{X}_s$. Two cases. (a) **SHORTSTOP** commits a certified chunk: Theorem 1 gives $\text{dist}(x_{t+1}, \mathcal{X}_u) \geq \epsilon$, and since $\mathcal{X}_s$ was chosen as $\{x : \text{dist}(x, \mathcal{X}_u) \geq \epsilon\}$ intersected with the robust control-invariant set, the one-step successor lies in $\mathcal{X}_s$ (the committed first action is exactly the input certified for step $k=1$ of the tube, whose reachable set is contained in the ϵ -safe region). (b) No admissible chunk exists and π_{fb} is applied: by Assumption 3, $f(x_t, \pi_{\text{fb}}(x_t)) \oplus \mathcal{B}(\bar{w}) \subseteq \mathcal{X}_s$, so $x_{t+1} \in \mathcal{X}_s$ under any admissible disturbance. In both cases $x_{t+1} \in \mathcal{X}_s$, completing the induction. Since $\mathcal{X}_s \cap \mathcal{X}_u = \emptyset$, the trajectory never enters $\mathcal{X}_u$. ■

C. Proof of Proposition 1

Inflation bound (6). Let r_k be the Hausdorff radius (worst half-width) of $\hat{\mathcal{R}}_k$. For $x, x' \in \hat{\mathcal{R}}_k$ and the same input a_{t+k} , $\|\hat{f}(x, a) - \hat{f}(x', a)\| \leq L\|x - x'\|$ by local Lipschitzness, so the image of a radius- r_k set has radius $\leq Lr_k$; Eq. (1) then adds $\varepsilon_{\max} + \bar{w}$, giving $r_{k+1} \leq Lr_k + (\varepsilon_{\max} + \bar{w})$. Unrolling this linear recursion yields $r_k \leq L^k r_0 + (\varepsilon_{\max} + \bar{w}) \sum_{j=0}^{k-1} L^j = L^k r_0 + (\varepsilon_{\max} + \bar{w}) \frac{L^k - 1}{L - 1}$ for $L \neq 1$, and the arithmetic sum $r_0 + k(\varepsilon_{\max} + \bar{w})$ for $L \leq 1$.

Rejection-rate bound (7). A chunk is certified-rejected iff its certified margin is below threshold, $\underline{\rho} < \epsilon$. Because $\underline{\rho}$ lower-bounds the true robustness by at most the tube radius (the interval bound loses at most r_H relative to the nominal value, since the tube's worst point is within r_H of the nominal trajectory), we have $\rho^{\text{nom}} - r_H \leq \underline{\rho} \leq \rho^{\text{nom}}$. Hence $\{\underline{\rho} < \epsilon\} \subseteq \{\rho^{\text{nom}} < \epsilon + r_H\}$, and

$$\alpha(H) = \Pr[\underline{\rho} < \epsilon] \leq \Pr[\rho^{\text{nom}} < \epsilon + r_H].$$

Split the event at 0: $\Pr[\rho^{\text{nom}} < 0] = \alpha^*$ (genuinely unsafe), plus the band $\Pr[0 \leq \rho^{\text{nom}} < \epsilon + r_H] \leq \beta(\epsilon + r_H)$ by the bounded-density assumption. Adding gives $\alpha(H) \leq \alpha^* + \beta(\epsilon + r_H)$.

Cost bound. Each certified rejection is resolved either by a fallback step (cost $\leq c_{\text{fb}}$) or by an admissible repair within the trust region (cost $\leq c_{\text{rep}}\delta$, since the executed chunk deviates from the policy sample by at most δ and g is c_{rep} -Lipschitz in that deviation). Charging the fraction $\alpha(H)$ of rejected decisions to a fallback and the executed repairs to their trust-region cost yields $\Delta J(H) \leq c_{\text{fb}}\alpha(H) + c_{\text{rep}}\delta$. ■

APPENDIX B

HYPERPARAMETERS AND PROTOCOL

Table VII lists all hyperparameters. Diffusion policies use a 1D-convolutional U-Net denoiser with 100 training / 10

inference DDIM steps and an action-chunk length of 16 (of which the shield certifies the first $H = 8$). Reachtubes use axis-aligned intervals for the six task-space safety coordinates (end-effector and elbow positions), with a 2nd-order zonotope refinement in the 2D study. The model-error radius ε is the 99th-percentile one-step residual on a held-out transition set, inflated by $1.25\times$. The counterexample search uses $M = 16$ projected-gradient steps over zonotope generators with random restarts at box vertices. The fallback is a critically-damped task-space braking controller with a pre-certified robust control-invariant stopping set.

TABLE VII
DEFAULT HYPERPARAMETERS.

Hyperparameter	Value
Candidate chunks K	8
Certified horizon H	8
CE-search iterations M	16
Safety margin ϵ	2 cm
Repair trust region δ	0.1
Repair step size η	0.05
Model-error quantile	99th ($\times 1.25$)
Disturbance bound $\bar{w}$	0.5 cm/step
Episodes $\times$ seeds	500×5
Bootstrap resamples	10^4